

# 1 Latency Comparison of Load Balancers

## 1.1 Custom Load Balancer Results

Table 1: Measured latency of custom load balancer

| Scenario                       | Avg Latency (ms) | p95 (ms) | p99 (ms) | RPS    |
|--------------------------------|------------------|----------|----------|--------|
| Part (b): All servers healthy  | 12.19            | 15.14    | 100.57   | 81.97  |
| Part (c): With server failures | 3.60             | 13.51    | 16.60    | 249.39 |

## 1.2 Request Distribution (Skew)

Table 2: Server request distribution (selected)

| Server   | Part (b) Requests | Part (c) Requests |
|----------|-------------------|-------------------|
| Server 1 | 7758              | 545               |
| Server 2 | 186               | 900               |
| Server 3 | 2050              | 9                 |
| Server 4 | 6                 | 8447              |
| Server 5 | 0                 | 0                 |
| Server 6 | 0                 | 0                 |
| Server 7 | 0                 | 0                 |
| Server 8 | 0                 | 0                 |

## 1.3 State-of-the-Art Load Balancers (Latency)

Table 3: Latency of production-grade load balancers

| Load Balancer       | Avg Latency (ms) | p95 (ms) | p99 (ms) |
|---------------------|------------------|----------|----------|
| HAProxy (HTTP)      | ~0.56            | –        | ~5       |
| NGINX (open source) | 101              | 302      | 846      |
| Traefik             | 91               | 237      | 687      |
| NGINX Plus          | <30              | –        | <13      |
| Envoy               | –                | ~855     | –        |

## 1.4 Comparative Summary

Table 4: Direct latency comparison

| System        | Avg (ms) | p95 (ms) | p99 (ms) |
|---------------|----------|----------|----------|
| Custom LB (b) | 12.19    | 15.14    | 100.57   |
| Custom LB (c) | 3.60     | 13.51    | 16.60    |
| HAProxy       | 0.56     | –        | 5        |
| NGINX Plus    | <30      | –        | <13      |

## 1.5 Conclusion

The custom load balancer achieves competitive average and p95 latency, and under failure conditions reaches p99 latency comparable to NGINX Plus. However, uneven request distribution causes tail latency degradation in healthy scenarios. Production load balancers such as HAProxy maintain lower and more stable tail latency through adaptive routing and advanced scheduling algorithms.